

Research Integrity: A Framework for Detection, Evidence & Remediation

Organised by unit of analysis: Individual → Team → Group/Collusion → Field/Community level

Problem / Driver	Evidence of Scope (key studies)	Detection Tools & Methods (year)	Prevention & Remediation (year)
Tier 1 Individual-level — fabrication, falsification, plagiarism, unintentional error			
Data fabrication <i>Single researcher invents raw data; often a junior team member or PI under grant/publication pressure</i>	▶ ORI annual reports (1992–): ~10 findings/yr on average ▶ Fang et al. (2012, PNAS): misconduct accounts for ~67% of medical retractions ▶ Stapel case: 58 papers retracted (2011) ▶ Gino case (Harvard 2024): 4 studies, data deliberately altered	▶ GRIM test (2016) ▶ SPRITE (2018) ▶ statcheck (2016) ▶ Image forensics / Bik (2014–) ▶ ORI investigation (1992–) ▶ False Claims Act settlements	▶ Open raw data sharing (2010s–) ▶ Lab notebooks & audit trails ▶ Data DOIs & versioning (Zenodo 2013, Figshare 2012) ▶ Federal whistleblower protections ▶ Bik Science Integrity Fund (2025)
Image manipulation <i>Duplicated, spliced, or AI-generated figures; increasingly hard to detect as generative AI matures</i>	▶ Bik et al. (2016, mBio): 3.8% of 20,000+ papers had problematic figures; ~50% showed signs of deliberate manipulation ▶ Imagetwin scan: 18–20% of papers in one journal had problem images (J. Cancer 2024–25) ▶ Tessier-Lavigne case: Stanford president resigned (2023)	▶ Bik visual screening (2014–) ▶ Imagetwin AI (early 2020s) ▶ Proofing AI (early 2020s) ▶ PubPeer crowdsourcing (2012–)	▶ Raw image submission required at peer review ▶ Figure metadata standards ▶ COPE retraction guidelines (2009–) ▶ Publisher AI screening at submission (2023–)
Plagiarism & text recycling <i>Copying text or results; self-plagiarism; AI-generated boilerplate now complicates detection</i>	▶ Cabanac et al. (2021): 2,000+ papers identified using 'tortured phrases' ▶ Nature (2025): 14% of biomedical abstracts showed AI-generated text markers	▶ iThenticate / CrossCheck (2008–) ▶ Tortured phrases screener (2021) ▶ AI-text detectors (2023–; reliability concerns)	▶ Mandatory plagiarism check at submission ▶ COPE self-plagiarism policies ▶ AI disclosure requirements at journals (2023–)
Research methods errors <i>Coding bugs, data entry errors, miscoded variables, mathematical errors; complexity scales error rates super-linearly</i>	▶ Reinhart & Rogoff (2010): Excel error reversed austerity-policy conclusion (Herndon et al. 2013) ▶ Eklund et al. (2016, PNAS): fMRI software bug affected ~40,000 studies ▶ Brodeur et al. (IZA 2023): coding errors in ~25% of economics papers re-executed ▶ Chang & Li (2022): only 14–43% of published econ studies computationally reproducible ▶ 75% of top 20 most-cited meta-analyses contained at least one error	▶ Re-execution / reproducibility checks ▶ PLOS ONE computational reproducibility (~2014) ▶ AEA Data Editor re-execution (2018–) ▶ Unit tests (software engineering) ▶ statcheck	▶ Code sharing mandates (NIH & NSF 2023) ▶ Code Ocean / Binder (2017–) ▶ Software citation standards (FORCE11 2016) ▶ Unit testing norms ▶ Gentzkow & Shapiro Code and Data guide (2014)
Tier 2a Team-level — analytic flexibility, p-hacking, HARKing, selective reporting, collaborative pipeline failure			
Collaborative pipeline failure <i>Errors emerge from handoffs between team members; one change cascades downstream; complexity scales errors super-linearly</i>	▶ NARPS (Botvinik-Nezer et al. 2020, Nature): 70 teams, same data → divergent conclusions on 5 of 9 hypotheses ▶ Gentzkow & Shapiro (2014): catalogue of real project failures ▶ Brodeur et al. (2023): pipeline failures common in reproducibility packages ▶ AlphaChip controversy (2024): data leakage, inconsistent environments	▶ Re-execution / push-button reproducibility tests ▶ Pull-request diff review ▶ targets / drake pipeline tools (R) ▶ Environment audit (renv, Docker) ▶ Data manifest / provenance logs	▶ Git / version control (~2012) ▶ Pull-request review workflow ▶ targets / drake pipeline tools (R, 2018–) ▶ renv / conda / Docker ▶ DVC (Data Version Control, 2019–) ▶ Snakemake / Nextflow (2012–) ▶ Code Ocean (2017–)
p-hacking / selective reporting	▶ John et al. (2012, Psych Sci): 58% of psychologists self-reported ≥1 QRP	▶ p-curve (Simonsohn et al. 2014) ▶ statcheck (2016)	▶ Preregistration (OSF 2013–) ▶ Registered Reports (2013–)

<i>Running many analyses, reporting only significant ones; stopping rules; covariate fishing; outcome switching</i>	▶ Simonsohn et al. (2014): p-curve revealed flat/left-skewed distributions in suspected sets ▶ Masicampo & Lalande (2012): suspicious pileup just below .05 in psychology journals ▶ Head et al. (2015, PLOS Bio): excess just-significant p-values across fields	▶ p-uniform (2015) ▶ Caliper test (Gerber & Malhotra 2008) ▶ z-curve 2.0 (Bartoš & Schimmack 2022) ▶ Registered vs. published outcome audit	▶ ClinicalTrials.gov outcome reporting (2000–) ▶ AllTrials campaign (2013)
HARKing <i>Presenting post-hoc hypotheses as a priori; rewriting introduction after seeing results; ~43% self-report doing this</i>	▶ Kerr (1998): coined the term; theoretical analysis of 12 costs ▶ Review of 6 surveys (2017): ~43% self-reported HARKing at least once ▶ O'Boyle et al. (2017): 'chrysalis effect' — hypotheses removed between dissertation and publication ▶ OSC (2015): lower replication rate for findings more likely to have been HARKed	▶ Compare registered vs. published hypotheses ▶ Preprint timestamp audit ▶ No automated tool — requires process evidence	▶ Preregistration (OSF 2013) ▶ Registered Reports — in-principle acceptance (2013–) ▶ Transparent changes protocol (OSF)
Fragile / researcher degrees-of-freedom findings <i>Result depends on arbitrary undisclosed analytic choices; technically honest but not robust; Gelman's 'garden of forking paths'</i>	▶ Silberzahn et al. (2018, AMPPS): 29 teams, same data — effect sizes ranged 0.89–2.93 ▶ Orben & Przybylski (2019): multiverse on screen time — effects ranged negative to positive ▶ Patel et al. (2015): vibration of effects — most findings sensitive to covariate choice	▶ Multiverse analysis (Steenen et al. 2016) ▶ Specification curve (Simonsohn et al. 2018) ▶ Vibration of effects (Patel et al. 2015) ▶ Sensitivity analyses	▶ Pre-specified analysis plans ▶ Robustness checks as field standard ▶ Open analysis code for reader re-specification ▶ FAIR data principles (2016)
Publication bias (team/PI decisions) <i>Teams selectively submit significant results; file-drawer problem; effect sizes inflated in published record</i>	▶ Sterling (1959): first documented — ~97% of published psychology results significant ▶ Franco et al. (2014, Science): 68% of null results from TESS experiments never written up ▶ Scheel et al. (2021): 96% of psychology papers report ≥1 significant finding	▶ Funnel plot / Egger test (1997) ▶ Trim-and-fill (Duval & Tweedie 2000) ▶ PET-PEESE meta-regression (2008) ▶ TES excess significance (Ioannidis 2007)	▶ Registered Reports (2013–) ▶ PLOS ONE outcomes-neutral review (2006) ▶ Journals of null results ▶ AllTrials / trial results reporting
Tier 2b Group / collusion-level — coordinated misconduct across teams, institutions, or journals			
Citation rings & uncritical peer review <i>Can be intentional or emergent; inflates citation metrics; hard to distinguish from legitimate scholarly community without formal analysis</i>	▶ Smaldino & McElreath (2016): model shows incentives naturally select for low-powered research ▶ Bibliometric studies in alternative medicine and some nutrition subfields (Ioannidis 2013) ▶ Documented cases in ego-depletion and power-posing literatures	▶ Citation network analysis ▶ Reviewer history transparency (open peer review) ▶ Bibliometric anomaly detection ▶ Co-citation clustering analysis	▶ Double-blind peer review ▶ COI disclosure requirements ▶ Cross-journal reviewer assignment ▶ Open peer review (F1000, eLife 2012–)
Fake peer review rings <i>Authors submit fake reviewer emails; paper mills arrange predetermined reviews; editors collude; accounts for ~50% of all retractions</i>	▶ Richardson & Amaral (PNAS 2025): 0.25% of PLOS ONE editors → 30.2% of retracted articles ▶ Scholarly Kitchen (2024): fake peer review now ~half of all retractions ▶ Wiley/Hindawi: 11,300+ papers retracted 2022–24	▶ Editor network analysis (PNAS 2025) ▶ Reviewer verification systems ▶ ORCID + institutional email validation ▶ Statistical anomaly detection by publishers	▶ Verified reviewer identity (ORCID 2012–) ▶ Publisher reviewer fraud detection (Wiley 2024) ▶ United2Act cross-publisher coordination ▶ Mass retraction protocols (COPE 2022–)
Paper mills & predatory journals <i>Industrial-scale fabrication; authorship for sale; APC-driven acceptance; journal hijacking; doubling every ~1.5 years</i>	▶ Richardson & Amaral (PNAS 2025): networks 'large, resilient and growing rapidly' ▶ Problematic Paper Screener: 32,786 suspected products identified (2024) ▶ Cancer research AI scan (2024): 14.4% of molecular oncology papers flagged	▶ Problematic Paper Screener (2021) ▶ Tortured phrases detection ▶ Hijacked Journal Checker (Retraction Watch) ▶ Beall's List (2010–) ▶ Image duplication clustering (PNAS 2025)	▶ DOAJ quality standards ▶ Think. Check. Submit. (2015) ▶ United2Act coordination ▶ Indexing requirements (PubMed, Scopus) ▶ COPE membership requirements

Institutional misconduct cover-up <i>Universities slow-walking investigations; protecting high-profile PIs; retaliating against whistleblowers; ORI enforcement declining</i>	▶ ORI (2025): only 2 findings — lowest since 2006; average ~10/yr ▶ Hauser case (Harvard): 3-year investigation, findings never disclosed publicly ▶ Tessier-Lavigne (Stanford): 8-month investigation only after external pressure	▶ ORI oversight (capacity declining) ▶ Investigative journalism (Science, Nature, Retraction Watch) ▶ False Claims Act settlements ▶ Congressional inquiries	▶ ORI mandatory investigation timelines ▶ Federal whistleblower protections ▶ Bik Science Integrity Fund (2025) ▶ Independent oversight proposals (Stockholm 2025)
Tier 3 Field / community-level — systemic biases, publication norms, incentive structures, data stewardship			
Publication bias at field level <i>Journals preferentially publish significant novel findings; null results in file drawer; inflated effect sizes across entire literatures</i>	▶ OSC Reproducibility Project (2015, Science): ~36–50% of 100 psychology findings replicated ▶ Camerer et al. (2018, Nature Human Behaviour): 62% replication rate in economics lab experiments ▶ z-curve meta-analyses (2022): estimated replication rate ~50–60% ▶ Ioannidis (2005): theoretical model — 'Why most published research findings are false'	▶ p-curve across literatures (2014) ▶ z-curve replication rate estimates (2022) ▶ TES excess significance (Ioannidis 2007) ▶ Bias-corrected meta-analysis (PET-PEESE, RMA)	▶ Registered Reports format (2013–) ▶ PLOS ONE outcomes-neutral review (2006) ▶ Replication journals (AMPPS 2018–) ▶ Pre-results peer review
Incentive structures: publish-or-perish & APC economics <i>Tenure tied to publication count and impact factor; NIH success rates ~19–20%; APC revenue model rewards volume over quality</i>	▶ John et al. (2012): QRP survey — career pressure correlated with self-reported QRPs ▶ NIH historical data: R01 success rates fell from ~31% (2000) to ~19% (2024) ▶ Elsevier profit: ~\$3.6B (2023) ▶ Fanelli (2010): proportion of 'positive' results increasing over time across disciplines	▶ Bibliometric audits ▶ Retraction rate by institution tracking ▶ QRP prevalence surveys (John et al. 2012) ▶ Fanelli-type trend analysis	▶ DORA declaration (2012) ▶ Leiden Manifesto (2015) ▶ Stockholm Declaration 34-point plan (2025) ▶ Diamond OA / subsidy models ▶ Quality-over-quantity hiring criteria
Lack of replication culture & incentives <i>No career credit for replication; novel findings rewarded over verification; failed replications hard to publish</i>	▶ Mueller-Langer et al. (2019): <0.1% of top 50 economics journal articles 1974–2014 were replications ▶ Makel, Plucker & Hegarty (2012): ~1% of top 100 psychology journal articles are replications ▶ Many Labs 1–4 (2013–2018): large variation in replication success	▶ OSC Reproducibility Project (2015) ▶ Many Labs projects (2013–2018) ▶ SCORE / DARPA replication programs (2019–) ▶ Institute for Replication (i4rep, 2022–)	▶ CRediT taxonomy (2014) ▶ Replication journals and special tracks (AMPPS 2018) ▶ NIH replication supplements ▶ Sleuths in Residence (Retraction Watch 2024) ▶ Big Team Science / Psych Science Accelerator (2017–)
Poor data stewardship & inaccessible research <i>Data lost after publication; no reproducible code; closed access; no persistent identifiers</i>	▶ Stodden et al. (2018, Science): only 26% of Science article authors shared data on request ▶ Gabelica et al. (2022): 79% of data in availability statements was unreachable ▶ Vines et al. (2014): data availability declines ~17% per year after publication	▶ Computational reproducibility checks ▶ Data availability statement audits ▶ FAIR compliance assessment tools (2021–) ▶ Curation & re-use failure tracking	▶ FAIR data principles (Wilkinson et al. 2016) ▶ NIH data sharing mandate (2023) ▶ NSF data management plans (2010–) ▶ Zenodo / Figshare / Dryad / OSF (2012–) ▶ Data DOIs & versioning (DataCite 2009–) ▶ Data availability statements (Nature 2016–) ▶ ORCID persistent researcher IDs (2012) ▶ RO-Crate research object packaging (2019–)
Reviewer burnout & declining peer review quality <i>Submission volume far exceeds reviewer capacity; superficial review enables errors and fraud; AI submissions accelerating</i>	▶ Publons Global State of Peer Review (2018): growing workload, declining willingness ▶ Brubaker (2023): ~15M peer review hours wasted annually on desk-rejected papers ▶ Nature (2023–): editors report dramatic increase in AI-generated manuscript submissions	▶ Open peer review audit trails ▶ Post-publication review (PubPeer 2012–) ▶ Review quality scoring (proposed)	▶ Open peer review (eLife, F1000 2012–) ▶ Reviewer recognition (Publons 2012) ▶ AI-assisted pre-screening (2023–) ▶ Portable peer review (Review Commons 2019) ▶ PREREview / open collaborative review

Sources cited in evidence column are primary empirical studies or systematic reviews. Full citations appear in the References section. Detection tools and remediation entries include approximate year of introduction.

References

Full bibliographic citations for empirical studies and reports referenced in the evidence, detection, and prevention columns of the framework table above. Listed alphabetically by first author.

- AllTrials Campaign (2013). All trials registered, all results reported. International advocacy initiative. www.alltrials.net
- Bartoš, F. & Schimmack, U. (2022). Z-curve 2.0: Estimating replication rates and discovery rates. *Meta-Psychology*, 6. <https://doi.org/10.15626/MP.2021.2720>
- Bik, E. M., Casadevall, A., & Fang, F. C. (2016). The prevalence of inappropriate image duplication in biomedical research publications. *mBio*, 7(3), e00809-16. <https://doi.org/10.1128/mBio.00809-16>
- Botvinik-Nezer, R., Holzmeister, F., Camerer, C. F., et al. (2020). Variability in the analysis of a single neuroimaging dataset by many teams. *Nature*, 582(7810), 84–88. <https://doi.org/10.1038/s41586-020-2314-9>
- Brodeur, A., Mikola, D., & Cook, N. (2023). Mass reproducibility and replicability: A new hope. IZA Discussion Paper No. 16912. <https://doi.org/10.2139/ssrn.4487320>
- Brown, N. J. L. & Heathers, J. A. J. (2017). The GRIM test: A simple technique detects numerous anomalies in the reporting of results in psychology. *Social Psychological and Personality Science*, 8(4), 363–369. <https://doi.org/10.1177/1948550616673876>
- Brubaker, M. (2023). Estimating peer review time costs in academic publishing. Scholarly Kitchen blog. <https://scholarlykitchen.sspnet.org>
- Cabanac, G., Labbé, C., & Magazinov, A. (2021). Tortured phrases: A dubious writing style emerging in science. Evidence of critical issues affecting established journals. *arXiv:2107.06751*.
- Camerer, C. F., Dreber, A., Holzmeister, F., et al. (2018). Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2, 637–644. <https://doi.org/10.1038/s41562-018-0399-z>
- Chang, A. C. & Li, P. (2022). Is economics research replicable? Sixty published papers from thirteen journals say 'usually not.' Finance and Economics Discussion Series 2015-083, Board of Governors of the Federal Reserve System. <https://doi.org/10.17016/FEDS.2015.083>
- San Francisco Declaration on Research Assessment (DORA). (2012). www.sfdora.org
- Duval, S. & Tweedie, R. (2000). Trim and fill: A simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics*, 56(2), 455–463. <https://doi.org/10.1111/j.0006-341X.2000.00455.x>
- Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *BMJ*, 315(7109), 629–634. <https://doi.org/10.1136/bmj.315.7109.629>
- Eklund, A., Nichols, T. E., & Knutsson, H. (2016). Cluster failure: Why fMRI inferences for spatial extent have inflated false-positive rates. *PNAS*, 113(28), 7900–7905. <https://doi.org/10.1073/pnas.1602413113>
- Fanelli, D. (2010). 'Positive' results increase down the hierarchy of the sciences. *PLOS ONE*, 5(4), e10068. <https://doi.org/10.1371/journal.pone.0010068>
- Fang, F. C., Steen, R. G., & Casadevall, A. (2012). Misconduct accounts for the majority of retracted scientific publications. *PNAS*, 109(42), 17028–17033. <https://doi.org/10.1073/pnas.1212247109>
- FORCE11. (2016). Software Citation Principles. *PeerJ Computer Science*, 2, e86. <https://doi.org/10.7717/peerj-cs.86>
- Franco, A., Malhotra, N., & Simonovits, G. (2014). Publication bias in the social sciences: Unlocking the file drawer. *Science*, 345(6203), 1502–1505. <https://doi.org/10.1126/science.1255484>
- Gabelica, M., Bojčić, R., & Puljak, L. (2022). Many researchers were not compliant with their published data sharing statement: mixed-methods study. *Journal of Clinical Epidemiology*, 150, 33–41. <https://doi.org/10.1016/j.jclinepi.2022.05.019>
- Gentzkow, M. & Shapiro, J. M. (2014). Code and data for the social sciences: A practitioner's guide. University of Chicago Booth School of Business. <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>
- Gerber, A. S. & Malhotra, N. (2008). Publication bias in empirical sociological research: Do arbitrary significance thresholds distort published results? *Sociological Methods & Research*, 37(1), 3–30. <https://doi.org/10.1177/0049124108318973>
- Head, M. L., Holman, L., Lanfear, R., Kahn, A. T., & Jennions, M. D. (2015). The extent and consequences of p-hacking in science. *PLOS Biology*, 13(3), e1002106. <https://doi.org/10.1371/journal.pbio.1002106>
- Heathers, J. A. J., Anaya, J., van der Zee, T., & Brown, N. J. L. (2018). Recovering data from summary statistics: Sample parameter reconstruction via iterative techniques (SPRITE). *PeerJ Preprints*. <https://doi.org/10.7287/peerj.preprints.26968v1>
- Herndon, T., Ash, M., & Pollin, R. (2014). Does high public debt consistently stifle economic growth? A critique of Reinhart and Rogoff. *Cambridge Journal of Economics*, 38(2), 257–279. <https://doi.org/10.1093/cje/bet075>
- Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLOS Medicine*, 2(8), e124. <https://doi.org/10.1371/journal.pmed.0020124>
- Ioannidis, J. P. A. & Trikalinos, T. A. (2007). An exploratory test for an excess of significant findings. *Clinical Trials*, 4(3), 245–253. <https://doi.org/10.1177/1740774507079441>
- John, L. K., Loewenstein, G., & Prelec, D. (2012). Measuring the prevalence of questionable research practices with incentives for truth telling. *Psychological Science*, 23(5), 524–532. <https://doi.org/10.1177/0956797611430953>
- Kerr, N. L. (1998). HARKing: Hypothesizing after the results are known. *Personality and Social Psychology Review*, 2(3), 196–217. https://doi.org/10.1207/s15327957pspr0203_4
- Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520, 429–431. <https://doi.org/10.1038/520429a>

- Makel, M. C., Plucker, J. A., & Hegarty, B. (2012). Replications in psychology research: How often do they really occur? *Perspectives on Psychological Science*, 7(6), 537–542. <https://doi.org/10.1177/1745691612460688>
- Masicampo, E. J. & Lalande, D. R. (2012). A peculiar prevalence of p values just below .05. *Quarterly Journal of Experimental Psychology*, 65(11), 2271–2279. <https://doi.org/10.1080/17470218.2012.711335>
- Mueller-Langer, F., Fecher, B., Harhoff, D., & Wagner, G. G. (2019). Replication studies in economics: How many and which papers are chosen for replication, and why? *Research Policy*, 48(1), 62–83. <https://doi.org/10.1016/j.respol.2018.07.019>
- Nuijten, M. B., Hartgerink, C. H. J., van Assen, M. A. L. M., Epskamp, S., & Wicherts, J. M. (2016). The prevalence of statistical reporting errors in psychology (1985–2013). *Behavior Research Methods*, 48, 1205–1226. <https://doi.org/10.3758/s13428-015-0664-2>
- O'Boyle, E. H., Banks, G. C., & Gonzalez-Mulé, E. (2017). The chrysalis effect: How ugly initial results metamorphosize into beautiful articles. *Journal of Management*, 43(2), 376–399. <https://doi.org/10.1177/0149206314527133>
- Orben, A. & Przybylski, A. K. (2019). The association between adolescent well-being and digital technology use. *Nature Human Behaviour*, 3, 173–182. <https://doi.org/10.1038/s41562-018-0506-1>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. <https://doi.org/10.1126/science.aac4716>
- Patel, C. J., Burford, B., & Ioannidis, J. P. A. (2015). Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations. *Journal of Clinical Epidemiology*, 68(9), 1046–1058. <https://doi.org/10.1016/j.jclinepi.2015.05.029>
- Reinhart, C. M. & Rogoff, K. S. (2010). Growth in a time of debt. *American Economic Review*, 100(2), 573–578. <https://doi.org/10.1257/aer.100.2.573>
- Richardson, R. A. K., Hong, S. S., Byrne, J. A., Stoeger, T., & Amaral, L. A. N. (2025). The entities enabling scientific fraud at scale are large, resilient, and growing rapidly. *PNAS*, 122(32), e2420092122. <https://doi.org/10.1073/pnas.2420092122>
- Scheel, A. M., Schijen, M. R. M. J., & Lakens, D. (2021). An excess of positive results: Comparing the standard psychology literature with registered reports. *Advances in Methods and Practices in Psychological Science*, 4(2). <https://doi.org/10.1177/25152459211007467>
- Silberzahn, R., Uhlmann, E. L., Martin, D. P., et al. (2018). Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science*, 1(3), 337–356. <https://doi.org/10.1177/2515245917747646>
- Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). P-curve: A key to the file-drawer. *Journal of Experimental Psychology: General*, 143(2), 534–547. <https://doi.org/10.1037/a0033242>
- Simonsohn, U., Simmons, J. P., & Nelson, L. D. (2020). Specification curve analysis. *Nature Human Behaviour*, 4, 1208–1214. <https://doi.org/10.1038/s41562-020-0912-z>
- Smaldino, P. E. & McElreath, R. (2016). The natural selection of bad science. *Royal Society Open Science*, 3(9), 160384. <https://doi.org/10.1098/rsos.160384>
- Steege, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing transparency through a multiverse analysis. *Perspectives on Psychological Science*, 11(5), 702–712. <https://doi.org/10.1177/1745691616658637>
- Sterling, T. D. (1959). Publication decisions and their possible effects on inferences drawn from tests of significance — or vice versa. *Journal of the American Statistical Association*, 54(285), 30–34. <https://doi.org/10.2307/2282137>
- Sabel, B. A. & Larhammar, D. (2025). Reformation of science publishing — the Stockholm Declaration. *Royal Society Open Science*, 12, 251805. <https://doi.org/10.1098/rsos.251805>
- Stodden, V., Seiler, J., & Ma, Z. (2018). An empirical analysis of journal policy effectiveness for computational reproducibility. *PNAS*, 115(11), 2584–2589. <https://doi.org/10.1073/pnas.1708290115>
- Vines, T. H., Albert, A. Y. K., Andrew, R. L., et al. (2014). The availability of research data declines rapidly with article age. *Current Biology*, 24(1), 94–97. <https://doi.org/10.1016/j.cub.2013.11.014>
- Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3, 160018. <https://doi.org/10.1038/sdata.2016.18>

Resources: Websites, Databases, Tools & Policy Documents

Active URLs verified April 2026. Links open in browser; some tools require free account registration.

Databases & Tracking

Retraction Watch Database — Searchable database of 63,000+ retracted papers with reason codes, maintained by the Center for Scientific Integrity. <https://retractionwatch.com>

PubPeer — Open post-publication peer review platform; search any paper by DOI for community annotations and flagged concerns. <https://pubpeer.com>

ORCID — Persistent digital identifiers for researchers; links publications, datasets, and review activity to a verified scholarly identity. <https://orcid.org>

DataCite — Persistent identifiers (DOIs) for research datasets; enables citation, versioning, and discovery of shared data. <https://datacite.org>

Detection Tools

statcheck (web app) — Paste a paper's text to automatically extract and recalculate reported p-values; flags inconsistencies. <https://statcheck.io>

p-curve app — Enter test statistics from a set of studies to run p-curve analysis; diagnoses evidential value vs. p-hacking. <https://www.p-curve.com>

Problematic Paper Screener — Automated screening tool that flags papers containing tortured phrases, image duplication signatures, and other paper-mill indicators. <https://www.irit.fr/~Guillaume.Cabanac/problematic-paper-screener>

Hijacked Journal Checker — Retraction Watch tool listing 400+ journals whose ISSN numbers have been hijacked by scam publishers.

<https://retractionwatch.com/the-retraction-watch-hijacked-journal-checker>

Imagetwin — AI-powered image integrity checking tool for publishers; detects duplicated or manipulated figures in submitted manuscripts.

<https://imagetwin.ai>

Science Integrity Digest (Elisabeth Bik) — Bik's blog documenting image manipulation cases; includes how-to guides for reporting concerns to journals.

<https://scienceintegritydigest.com>

Open Science Infrastructure

Open Science Framework (OSF) — Free platform for preregistration, preprints, data and code sharing, and collaborative project management.

<https://osf.io>

Zenodo — Free, open-access repository for research data, software, and other outputs; hosted by CERN; issues DOIs. <https://zenodo.org>

Figshare — Online repository for sharing research data, figures, and other outputs; issues persistent identifiers. <https://figshare.com>

Dryad — General-purpose data repository focused on scientific and medical research; integrates with many journal submission systems.

<https://datadryad.org>

Code Ocean — Cloud-based platform for publishing reproducible research code; every compute capsule gets a persistent DOI. <https://codeocean.com>

Review Commons — Platform offering portable peer review for life science manuscripts; a single review transfers across 17 affiliate journals.

<https://www.reviewcommons.org>

Journals, Publishing Standards & Ethics Bodies

COPE — Committee on Publication Ethics — International body providing guidelines, flowcharts, and case discussions on editorial ethics; membership for publishers and journals. <https://publicationethics.org>

DOAJ — Directory of Open Access Journals — Quality-controlled index of open-access journals; inclusion signals adherence to transparency and peer review standards. <https://doaj.org>

Think. Check. Submit. — Checklist helping researchers evaluate journal legitimacy before submission; available in 25+ languages.

<https://thinkchecksubmit.org>

DORA — Declaration on Research Assessment — International declaration urging institutions to stop using journal impact factor in hiring, promotion, and funding decisions. <https://sfedora.org>

Leiden Manifesto — Ten evidence-based principles for the responsible use of quantitative research metrics; published in Nature (2015).

<https://www.leidenmanifesto.org>

Regulatory, Legal & Oversight Bodies

US Office of Research Integrity (ORI) — Federal body overseeing misconduct in NIH/HHS-funded research; issues findings, sanctions, and debarment decisions. <https://ori.hhs.gov>

False Claims Act (31 U.S.C. §§ 3729–3733) — Federal law enabling private parties ('relators') to sue on behalf of the government for fraudulent use of federal research funds; qui tam provisions allow whistleblowers to share in recovered damages. <https://www.justice.gov/civil/false-claims-act>

ClinicalTrials.gov — NIH registry for clinical trials; mandatory registration of primary outcomes before unblinding and results reporting within 12 months of completion. <https://clinicaltrials.gov>

AllTrials Campaign — International advocacy initiative calling for full public reporting of all clinical trial results — positive, null, and negative.

<https://www.alltrials.net>

Guides & Educational Resources

Gentzkow & Shapiro: Code and Data for the Social Sciences — Practitioner's guide covering automation, version control, directory management, abstraction, documentation, and team coordination for empirical research. <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>

Preregistration (OSF guide) — Step-by-step guidance on preregistering a study on the Open Science Framework; templates for many study types.

<https://help.osf.io/article/145-preregistration>

FAIR Data Principles — The original Wilkinson et al. (2016) paper defining Findability, Accessibility, Interoperability, and Reusability for scientific data.

<https://www.go-fair.org/fair-principles>

COPE Retraction Guidelines — Publisher guidance on when and how to issue corrections, expressions of concern, and retractions; downloadable flowcharts. <https://publicationethics.org/retraction-guidelines>

Stockholm Declaration (2025) — 34-point action plan for reforming science publishing and research assessment; published in Royal Society Open Science.

<https://doi.org/10.1098/rsos.251805>

Elisabeth Bik Science Integrity Fund — Fund supporting volunteer sleuths and science integrity advocates; managed in partnership with the Center for Scientific Integrity. <https://centerforscientificintegrity.org/the-elisabeth-bik-science-integrity-fund>

Registered Reports: explanation and list of participating journals — Over 300 journals now offer the Registered Reports format (peer review before data collection); Center for Open Science maintains the current list. <https://cos.io/rr>

